

Introduction to Android Apps

Day 01 - Session 01

21st September 2026

C-DAC, Hyderabad

01 Introduction To Mobile Apps

02 The App Sandbox and the Permission Model

03 Android Security Architecture

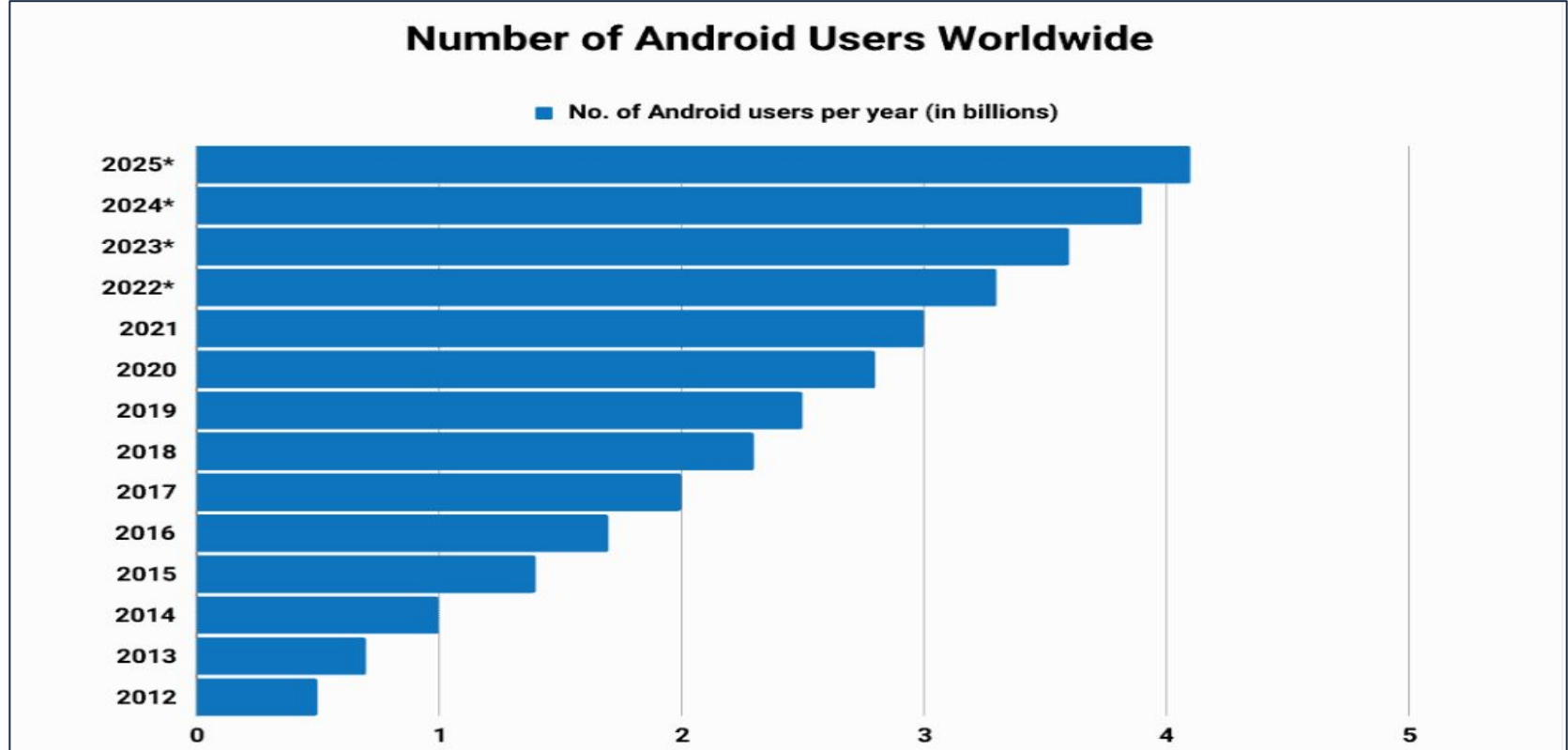
04 Android File System

Introduction To Mobile Apps

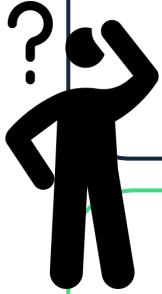
- **Smartphones are now everyday essentials.**
- **They have transformed communication, internet use, payments, gaming, and more.**
- **India: 1B+ smartphone users by 2026; 1.55B by 2040.**
- **Smartphones are driving digital growth in India.**



Introduction To Mobile Apps



Introduction To Mobile Apps



What is Android?

- Mobile OS for smartphones & tablets
- Developed by Google
- Used by ~ 72% of mobile devices worldwide

Why is Android Popular?

- Open-source (free to use)
- Customizable and flexible
- Supports a wide range of apps
- Affordable devices

Introduction To Mobile Apps

Key Features of Android



Touchscreen-friendly interface



Google Play Store with millions of apps



Regular updates for new features and security



Works on various devices (Phones, Tablets, TVs and Cars)

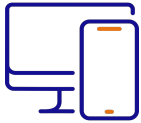
Android Development Tools



Android Studio, IntelliJ IDEA, VS Code



Android Software Development Kit (SDK) (API 34, 35, 36)



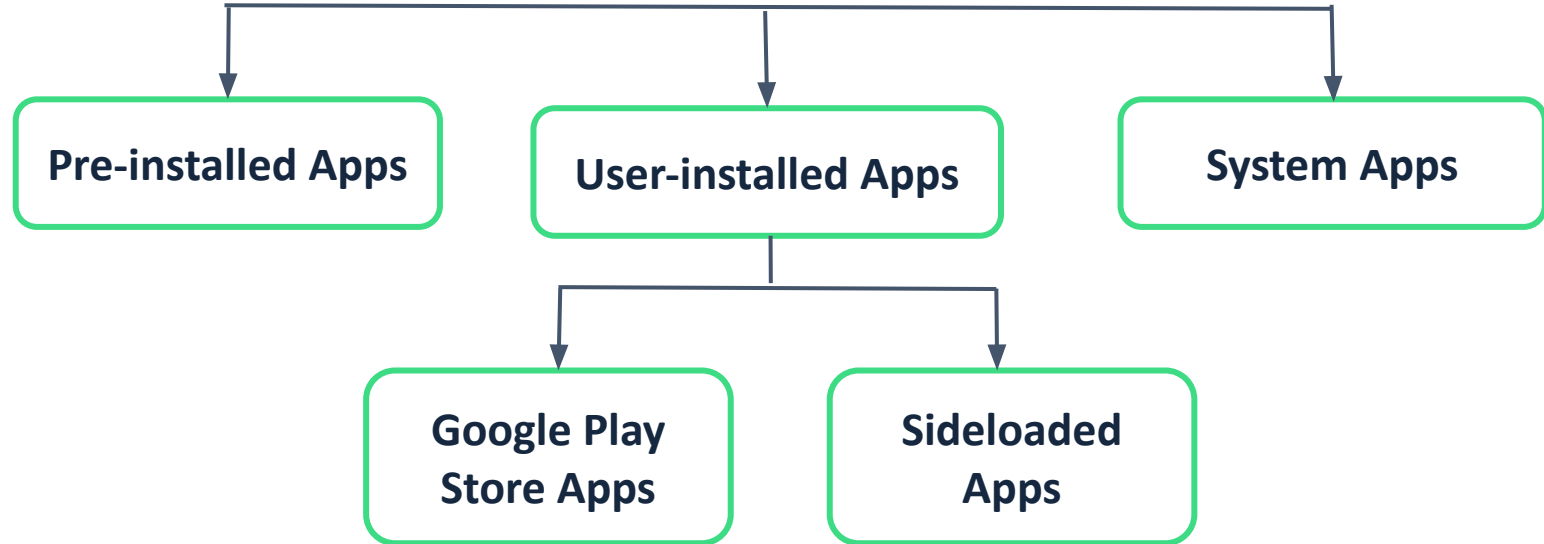
Emulators and Physical Devices for Testing (BrowserStack, Genymotion)



Debugging and Profiling Tools (Android Studio Debugger, Logcat / ADB , Android Studio Profilers)

Introduction To Mobile Apps

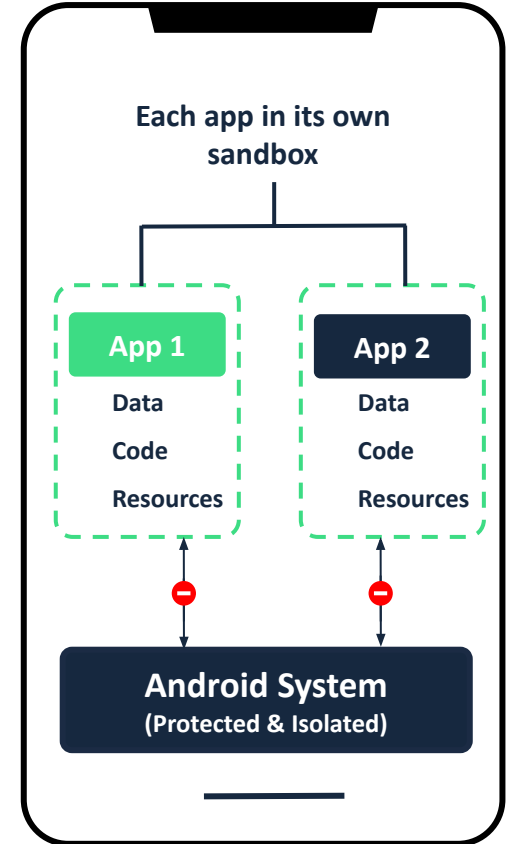
Types of Android applications



The App Sandbox And The Permission Model

The Application Sandbox

- App sandboxing is a security mechanism that isolates each app's data, code, and resources from other apps and the underlying Android system by enforcing strict security boundaries.
- This isolation prevents malicious apps from accessing sensitive information or performing unauthorized actions on the device.



The Permission Model

- Android's permission system governs access to sensitive resources and user data by apps.
- Permissions are declarations in the app's manifest file that specify what resources or actions an app can access.
- The permission system ensures that apps operate within defined boundaries and prevents unauthorized access to sensitive information.

Types of Permissions

Normal

Low-risk access with minimal impact on privacy
Ex: INTERNET

Signature

Granted only to apps signed with the same certificate as the defining app

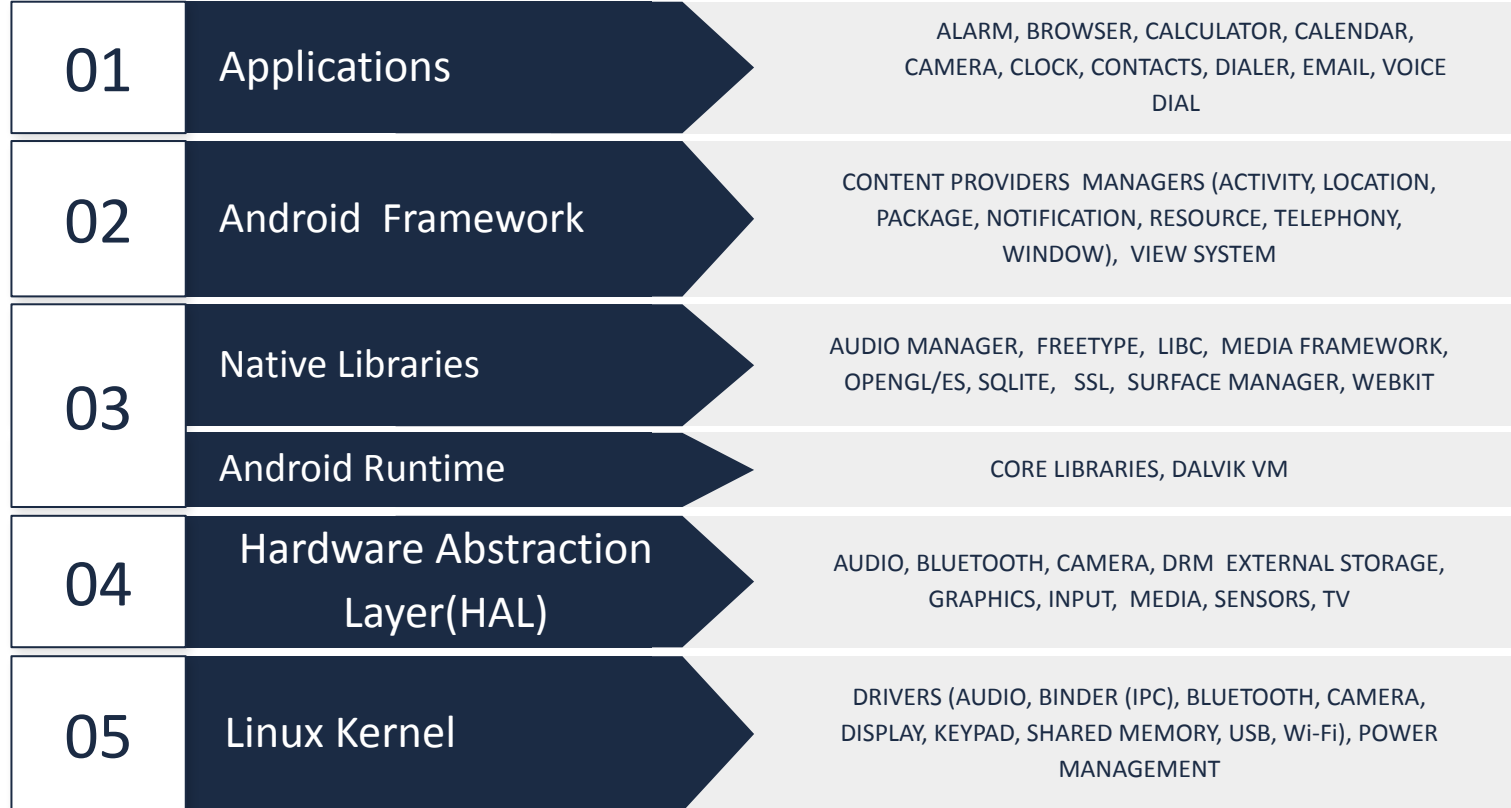
Runtime

Requires explicit user approval while the app is running
Ex: CAMERA

Special

Provides access to powerful operations through special system settings
Ex: SYSTEM_ALERT_WINDOW

Android Security Architecture



Android File System

Android File System

Android uses a file system similar to the disk-based file systems found on other platforms.



App-Specific Storage

Store files intended only for your application in dedicated internal or external storage directory



Shared Storage

Store files that need to be accessible or shared with other applications



Shared Preferences

Store small, private data as key-value pairs



Databases

Store structured application data in a private database

Android File System

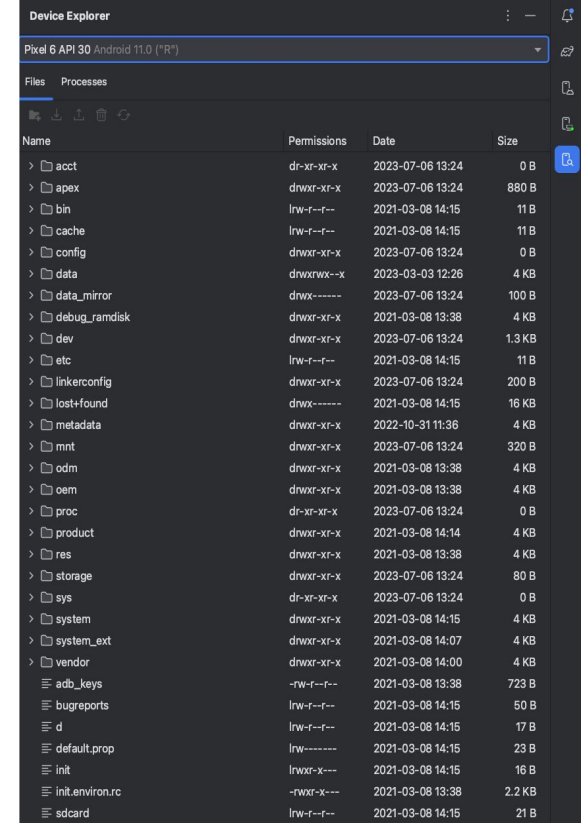
Categories of Storage Locations

Internal Storage

External Storage

Permissions and access to external storage

- READ_MEDIA_IMAGES
- READ_MEDIA_VIDEO
- READ_MEDIA_AUDIO



The screenshot shows the 'Device Explorer' window in Android Studio, displaying the file system of a 'Pixel 6 API 30 (R*)' virtual device. The 'Files' tab is selected, showing a list of system directories and files with their permissions, dates, and sizes.

Name	Permissions	Date	Size
> acct	dr-xr-xr-x	2023-07-06 13:24	0 B
> apex	drwxr-xr-x	2023-07-06 13:24	880 B
> bin	lrwxr--r--	2021-03-08 14:15	11 B
> cache	lrwxr--r--	2021-03-08 14:15	11 B
> config	drwxr-xr-x	2023-07-06 13:24	0 B
> data	drwxrwx--x	2023-03-03 12:26	4 KB
> data_mirror	drwx-----	2023-07-06 13:24	100 B
> debug_ramdisk	drwxr-xr-x	2021-03-08 13:38	4 KB
> dev	drwxr-xr-x	2023-07-06 13:24	1.3 KB
> etc	lrwxr--r--	2021-03-08 14:15	11 B
> linkerconfig	drwxr-xr-x	2023-07-06 13:24	200 B
> lost+found	drwx-----	2021-03-08 14:15	16 KB
> metadata	drwxr-xr-x	2022-10-31 11:36	4 KB
> mint	drwxr-xr-x	2023-07-06 13:24	320 B
> odm	drwxr-xr-x	2021-03-08 13:38	4 KB
> oem	drwxr-xr-x	2021-03-08 13:38	4 KB
> proc	dr-xr-xr-x	2023-07-06 13:24	0 B
> product	drwxr-xr-x	2021-03-08 14:14	4 KB
> res	drwxr-xr-x	2021-03-08 13:38	4 KB
> storage	drwxr-xr-x	2023-07-06 13:24	80 B
> sys	dr-xr-xr-x	2023-07-06 13:24	0 B
> system	drwxr-xr-x	2021-03-08 14:15	4 KB
> system_ext	drwxr-xr-x	2021-03-08 14:07	4 KB
> vendor	drwxr-xr-x	2021-03-08 14:00	4 KB
≡ adb_keys	-rw-r--r--	2021-03-08 13:38	723 B
≡ bugreports	lrwxr--r--	2021-03-08 14:15	50 B
≡ d	lrwxr--r--	2021-03-08 14:15	17 B
≡ default.prop	lrwx-----	2021-03-08 14:15	23 B
≡ init	lrwxr-x---	2021-03-08 14:15	16 B
≡ init.environ.rc	-rwxr-x---	2021-03-08 13:38	2.2 KB
≡ sdcard	lrwxr--r--	2021-03-08 14:15	21 B

THANK YOU!!